



LECTURE #10

Understanding Raster Data

REVIEW QUESTIONS

1. WHAT IS RASTER DATA AND HOW IS IT DIFFERENT FROM VECTOR DATA?
2. WHAT DOES THE TERM SPATIAL RESOLUTION MEAN IN A SATELLITE IMAGE?
3. HOW DO SPECTRAL BANDS HELP US ANALYZE THE EARTH'S SURFACE?
4. WHICH BANDS ARE USED TO CREATE A TRUE COLOR IMAGE (RGB)?
5. WHY ARE SENTINEL-2 AND LANDSAT IMPORTANT MISSIONS FOR ENVIRONMENTAL STUDIES?
6. HOW CAN WE ACCESS AND DOWNLOAD FREE SATELLITE IMAGERY FROM COPENICUS DATA SPACE?
7. WHAT IS THE PURPOSE OF THE .SAFE STRUCTURE IN SENTINEL-2 PRODUCTS?

TO THINK ABOUT...

IF YOU HAD TO EXPLAIN WHAT A RASTER IS TO A FRIEND IN ONE SENTENCE,
HOW WOULD YOU DESCRIBE IT?



LECTURE #11

Processing Raster Data

OBJECTIVE

- OPEN, CLIP AND CALCULATE SPECTRAL INDICES TO ANALYSE WATER AND VEGETATION USING SENTINEL IMAGES (10M RESOLUTION).

STEPS

1. OPEN AND CLIP IMAGES.
2. UNDERSTAND NDVI (VEGETATION INDEX)
3. PROCESS NDVI USING QGIS
4. UNDERSTAND NDWI (WATER INDEX)
5. PROCESS NDWI USING QGIS

1 – ADD RASTER TO QGIS

THE MOST SIMPLE WAY TO ADD A RASTER TO QGIS IS:

MENU LAYER → ADD LAYER → ADD RASTER LAYER

YOU CAN ALSO ACTIVATE THE BROWSER PANEL TO
BETTER VISUALIZE YOUR ‘.SAFE’ FILE (CONTAINING
SENTINEL DATA)

MENU VIEW → PANELS → BROWSER

2 – CLIP RASTER

TO CLIP RASTERS FROM ‘.SAFE’ FILE, WE NEED
TO:

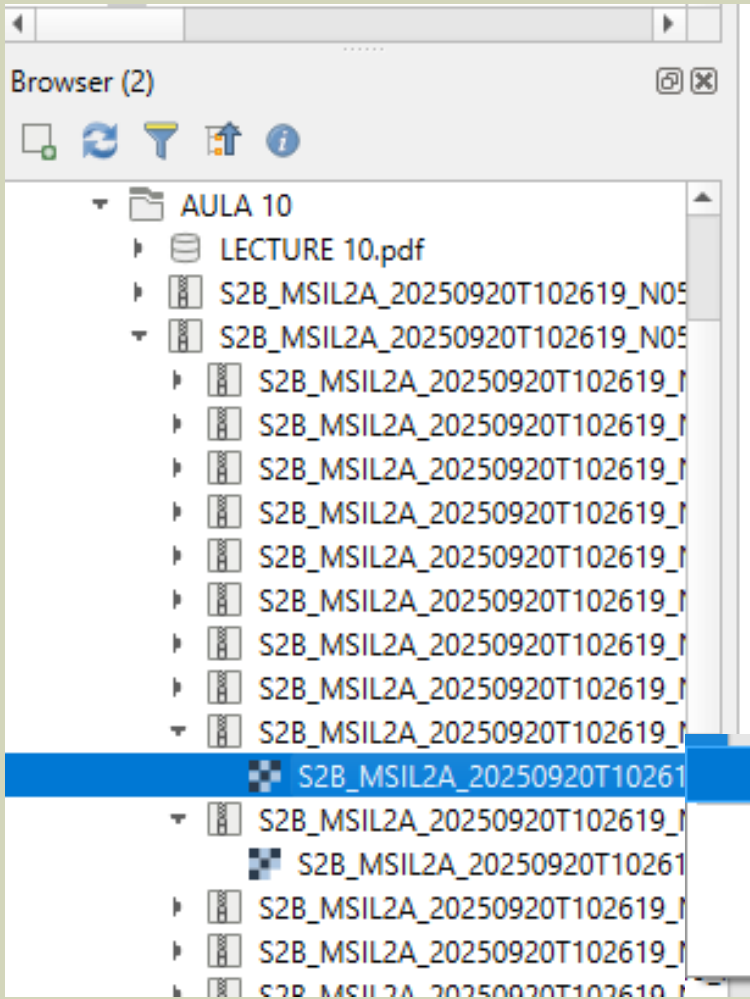
- EXTRACT THE RASTER FROM THE .SAFE FILE
- CLIP RASTER: MENU RASTER → EXTRACTION

EXAMPLE (ADDING AND CLIPPING)

LET’S OPEN ALL BANDS AVAILABLE FOR 10M

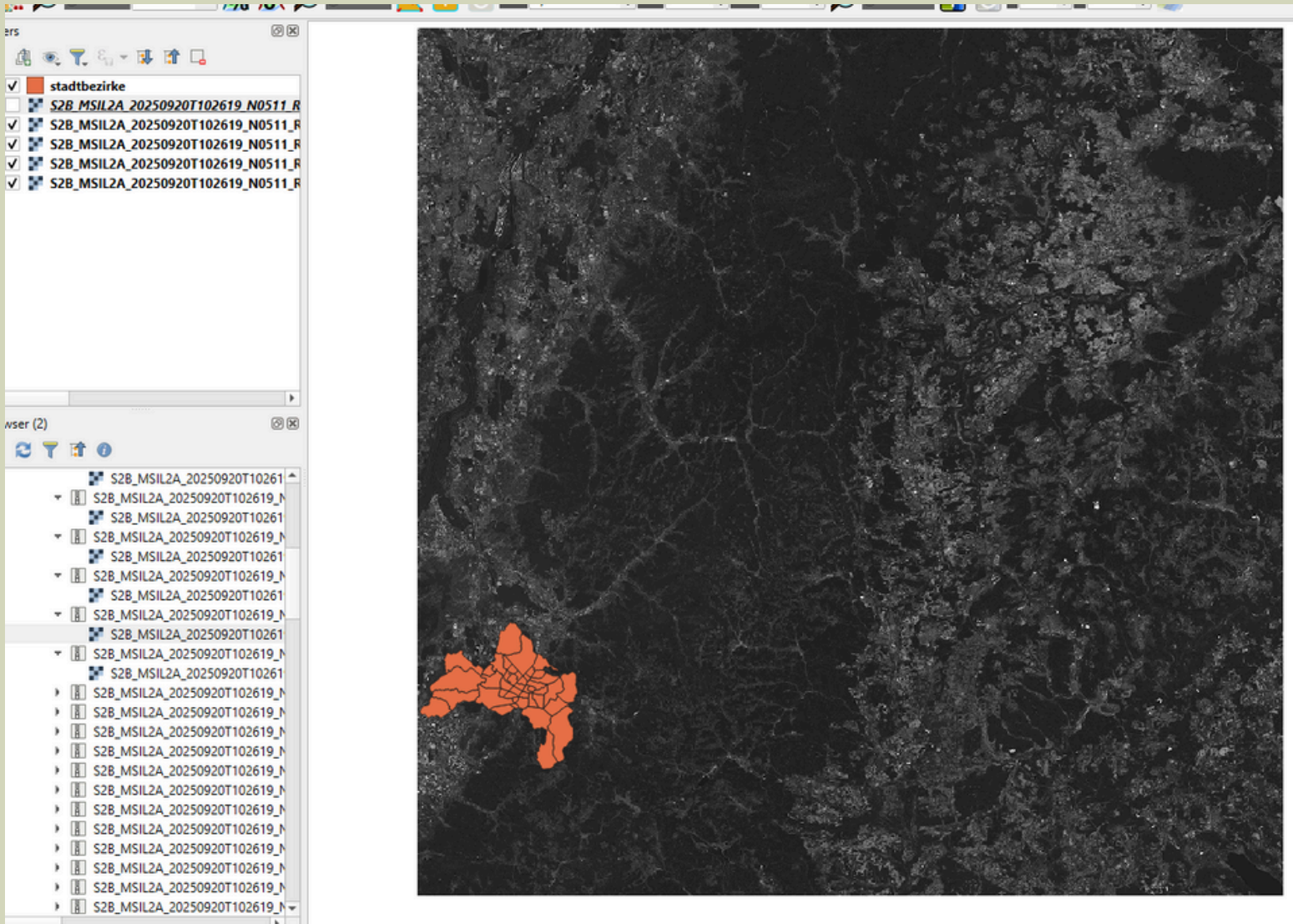
RESOLUTION:

1. SEARCH IN BROWSER AND CLICK 2X TO ADD TO QGIS



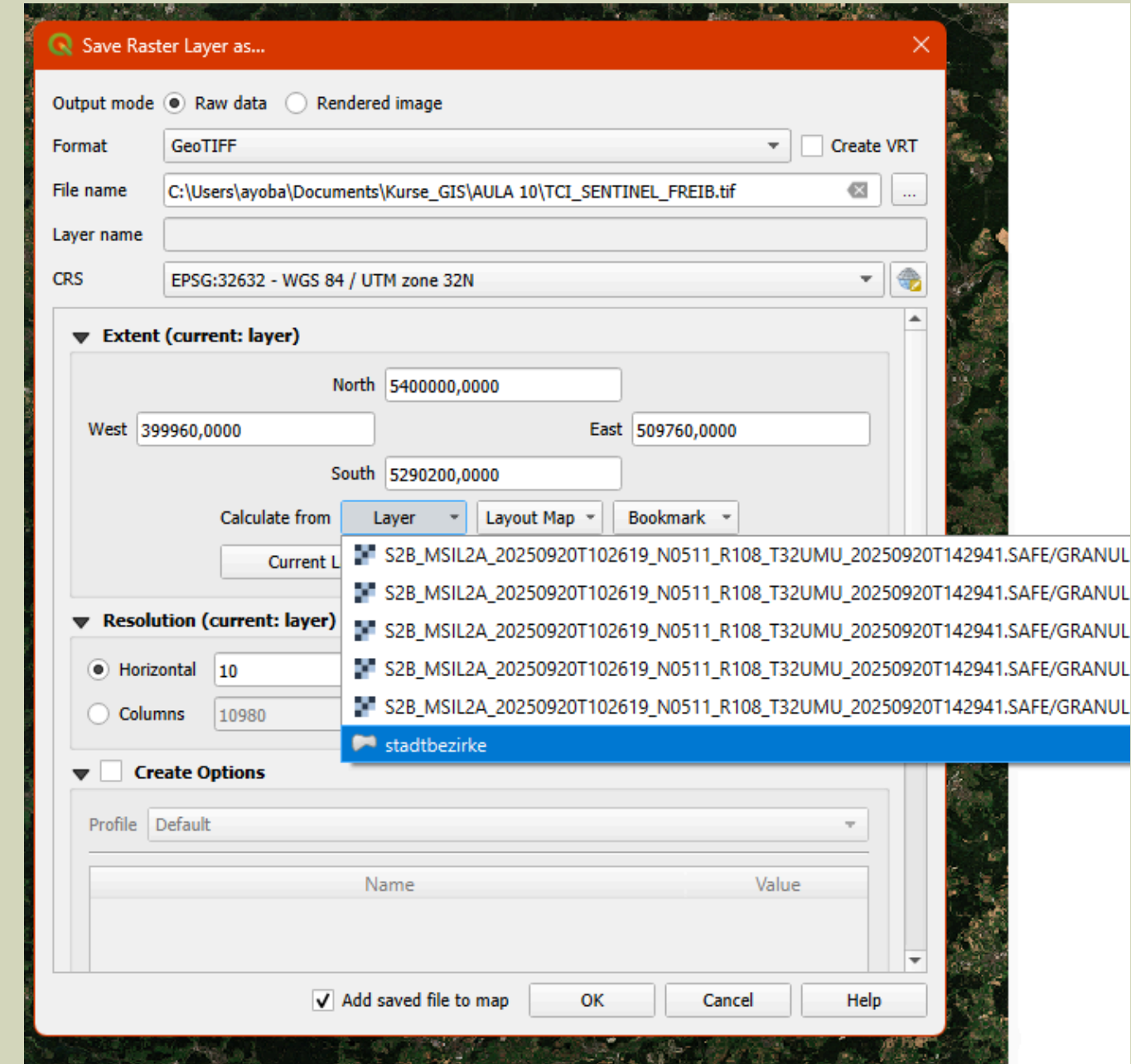
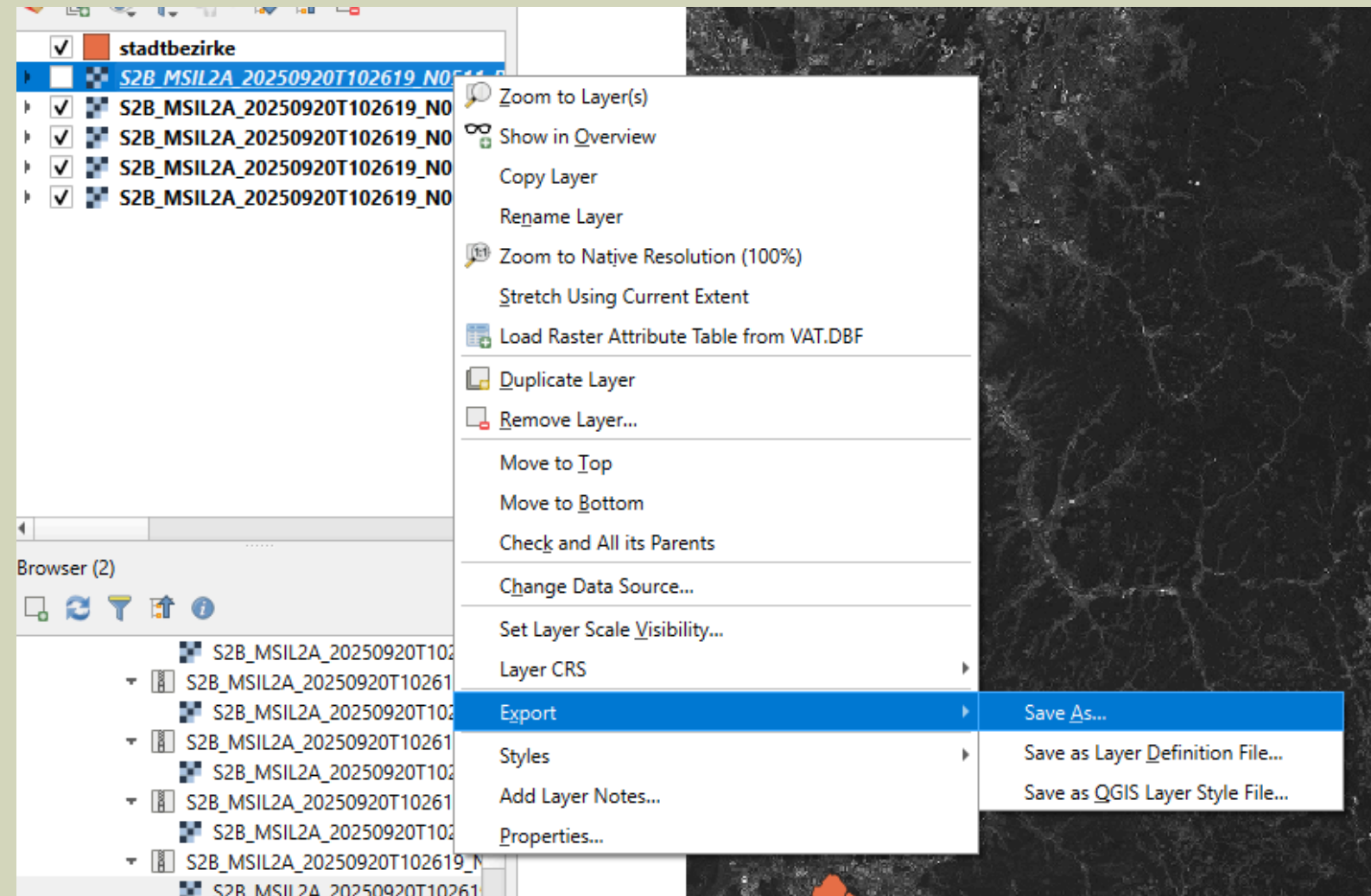
Band	Nome
B02	Blue (Azul)
B03	Green (Verde)
B04	Red (Vermelho)
B08	NIR (Near Infrared)

2. ADD THE VECTOR LAYER OF BREIBURG’S NEIGHBORHOODS.



EXAMPLE (ADDING AND CLIPPING)

3. EXTRACT AND CLIPP



EXAMPLE (ADDING AND CLIPPING)

RESULT: FIVE IMAGES CLIPPED TO FREIBURG'S
EXTENT (SQUARE)

- ✓  TCI_SENTINEL_FREIB
- ✓  sentinel_B4_Freib
- ✓  sentinel_B8_Freib
- ✓  sentinel_B2_Freib
- ✓  sentinel B3 Freib



3 – ANALIZING VEGETATION WITH NDVI

VEGETATION INTERACTS WITH SUNLIGHT IN A VERY SPECIFIC AND PREDICTABLE WAY.

ESPECIALLY IN **THE RED (RED)** AND **NEAR-INFRARED (NIR)** WAVELENGTHS.

- HEALTHY VEGETATION STRONGLY ABSORBS RED LIGHT. (RED REFLECTANCE IS LOW OVER HEALTHY PLANTS).
- LEAVES STRONGLY REFLECT NIR DUE TO THEIR INTERNAL CELL STRUCTURE.
 - SOIL, WATER, AND BUILT SURFACES DO NOT HAVE THIS INTERNAL CELLULAR STRUCTURE, SO THEY REFLECT VERY LITTLE NIR.

**NORMALIZED
DIFFERENCE
VEGETATION
INDEX**

IS USED IN REMOTE SENSING TO MEASURE THE **HEALTH** AND **DENSITY** OF VEGETATION.

NDVI COMPARES NIR AND RED CONSIDERING:

- HEALTHY VEGETATION → HIGH NIR AND LOW RED
- STRESSED VEGETATION → NIR DECREASES + RED INCREASES
- WATER/SOIL → BOTH ARE LOW → RESULTS CLOSE TO ZERO OR NEGATIVE

$$NDVI = \frac{NIR - RED}{NIR + RED}$$

NDVI WORKS BECAUSE HEALTHY PLANTS ABSORB RED LIGHT FOR PHOTOSYNTHESIS AND REFLECT A LOT OF NEAR-INFRARED LIGHT DUE TO THE INTERNAL STRUCTURE OF THE LEAF. THE STRONGER THIS DIFFERENCE, THE HIGHER THE NDVI VALUE, INDICATING VIGOROUS VEGETATION.

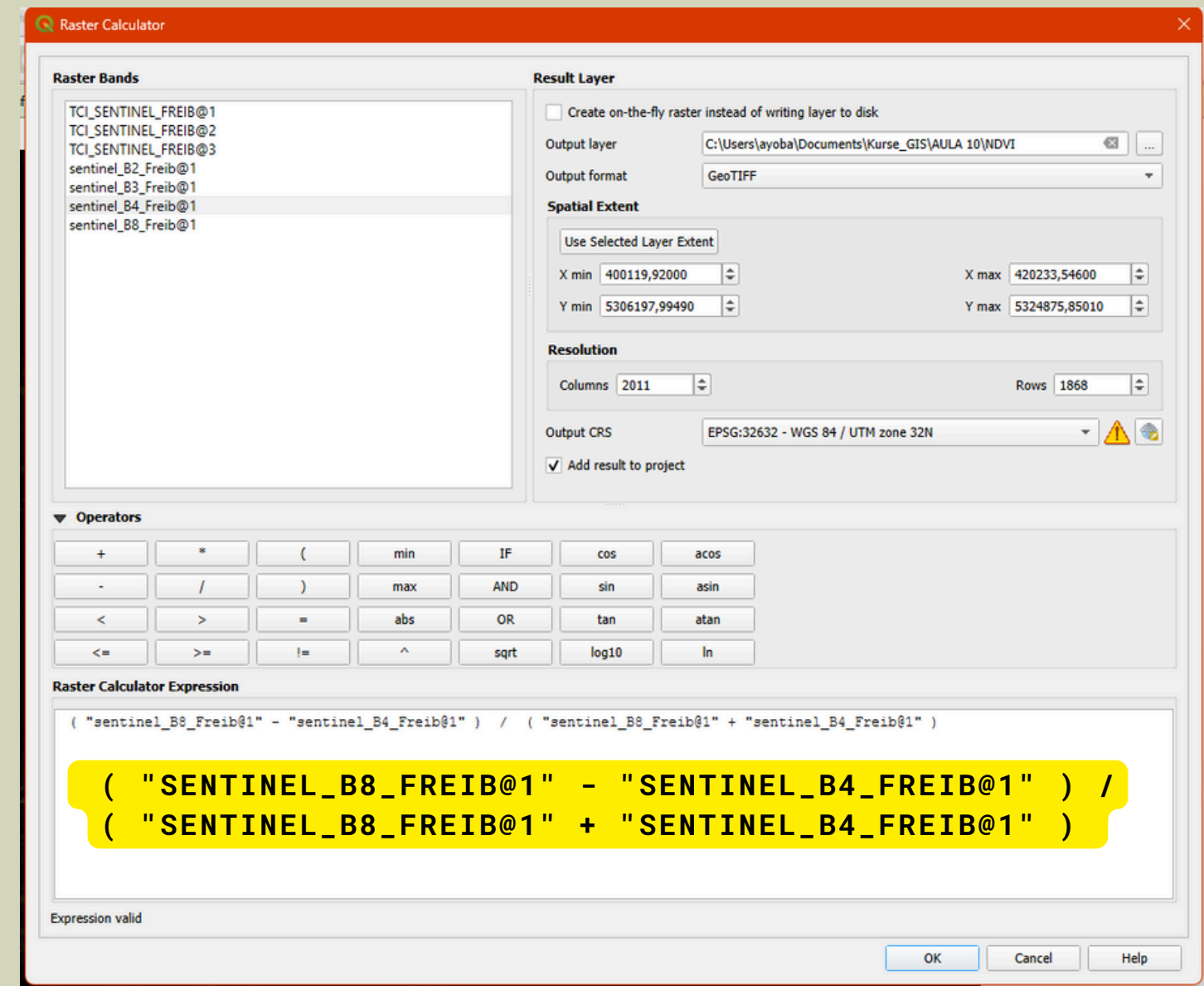
4 – PROCESS NDVI IN QGIS

$$NDVI = \frac{NIR - RED}{NIR + RED}$$

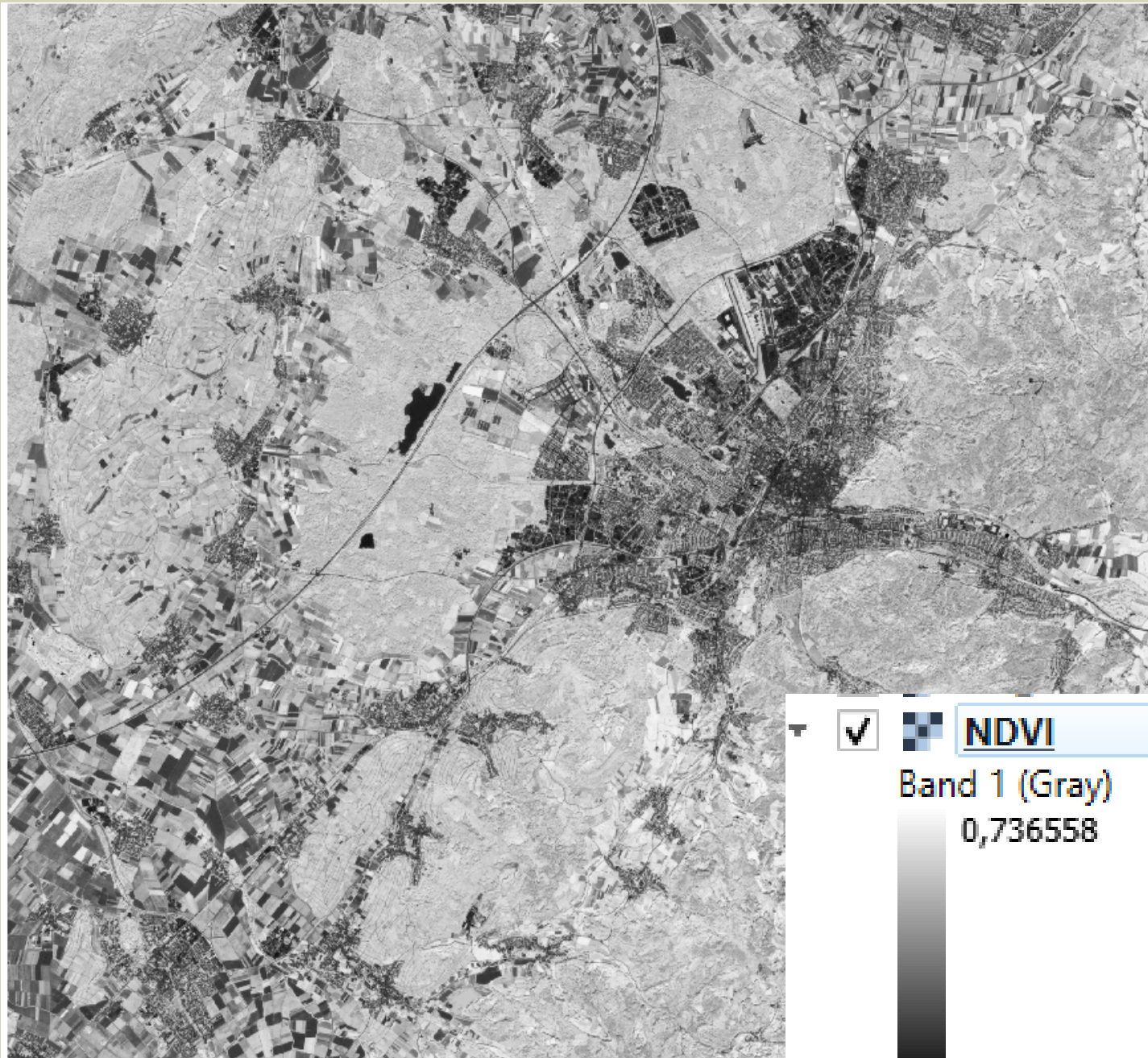
BAND	NAME
B02	Blue (Azul)
B03	Green (Verde)
B04	Red (Vermelho)
B08	NIR (Near Infrared)

$$NDVI = \frac{B8 - B4}{B8 + B4}$$

MENU RASTER → RASTER CALCULATOR



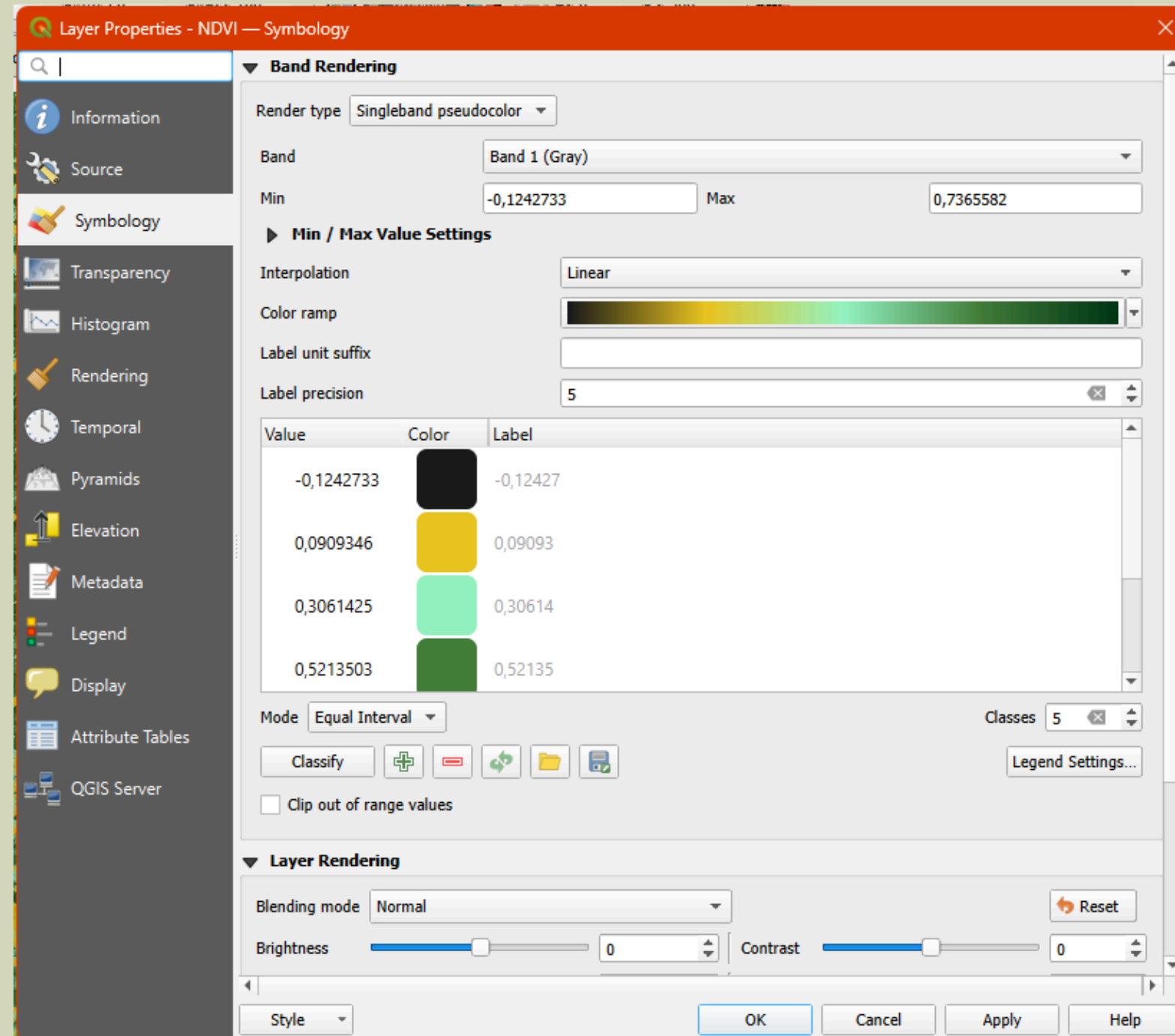
4 – PROCESS NDVI IN QGIS



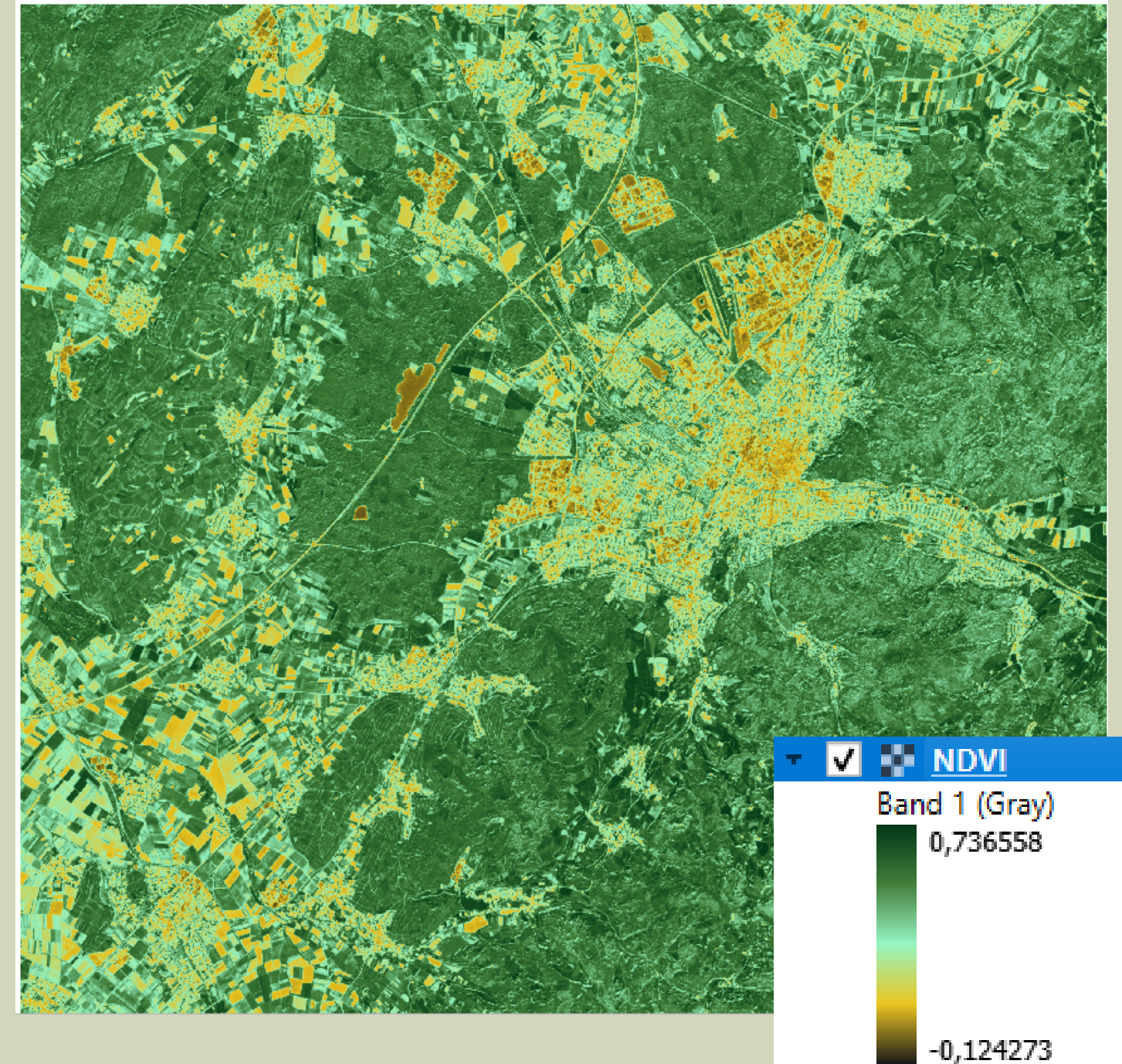
NDVI value	Meaning / Interpretation
0.6 to 1.0	Dense, healthy vegetation (forest, vigorous crops)
0.4 to 0.6	Moderate, healthy vegetation (grasslands, agriculture)
0.2 to 0.4	Sparse or stressed vegetation
0.1 to 0.2	Bare soil with little vegetation
0.0 to 0.1	Rock, sand, urban surfaces, built-up areas
< 0	Water, shadows, clouds, very dark surfaces

YOU CAN CHANGE THE SYMBOLOGY
PROPERTIES → SYMBOLOGY

3 – PROCESS NDVI IN QGIS



YOU CAN CHANGE THE SYMBOLOGY
PROPERTIES → SYMBOLOGY



5 – ANALIZING WATER WITH NDWI

09

1. WATER ABSORBS NEAR-INFRARED (NIR) LIGHT
 - WATER ABSORBS ALMOST ALL NIR RADIATION.
 - THIS MAKES NIR REFLECTANCE VERY LOW OVER BODIES OF WATER.
2. WATER REFLECTS A LITTLE GREEN LIGHT
 - WATER REFLECTS SOME GREEN WAVELENGTHS.
 - THIS REFLECTION IS STILL SMALL, BUT GREATER THAN NIR.
3. VEGETATION AND SOIL BEHAVE DIFFERENTLY
 - PLANTS REFLECT A LOT OF NIR AND ABSORB GREEN.
 - SOIL AND BUILT SURFACES REFLECT BOTH GREEN AND NIR AT SIMILAR LEVELS.

THIS DIFFERENCE CREATES A STRONG CONTRAST
 WATER → GREEN > NIR
 VEGETATION → NIR >> GREEN
 SOIL → GREEN ≈ NIR

○

NORMALIZED DIFFERENCE WATER INDEX

USED TO IDENTIFY AND
 MAP SURFACE WATER,
 SUCH AS RIVERS,
 LAKES, RESERVOIRS,
 PONDS, AND FLOODED
 AREAS.

NDWI COMPARES HOW MUCH GREEN LIGHT AND NEAR-
 INFRARED LIGHT (NIR) ARE REFLECTED FROM THE
 EARTH'S SURFACE.

$$NDWI = \frac{\text{Green} - \text{NIR}}{\text{Green} + \text{NIR}}$$

NDWI WORKS BECAUSE WATER REFLECTS LIGHT VERY
 DIFFERENTLY FROM SOIL AND VEGETATION.

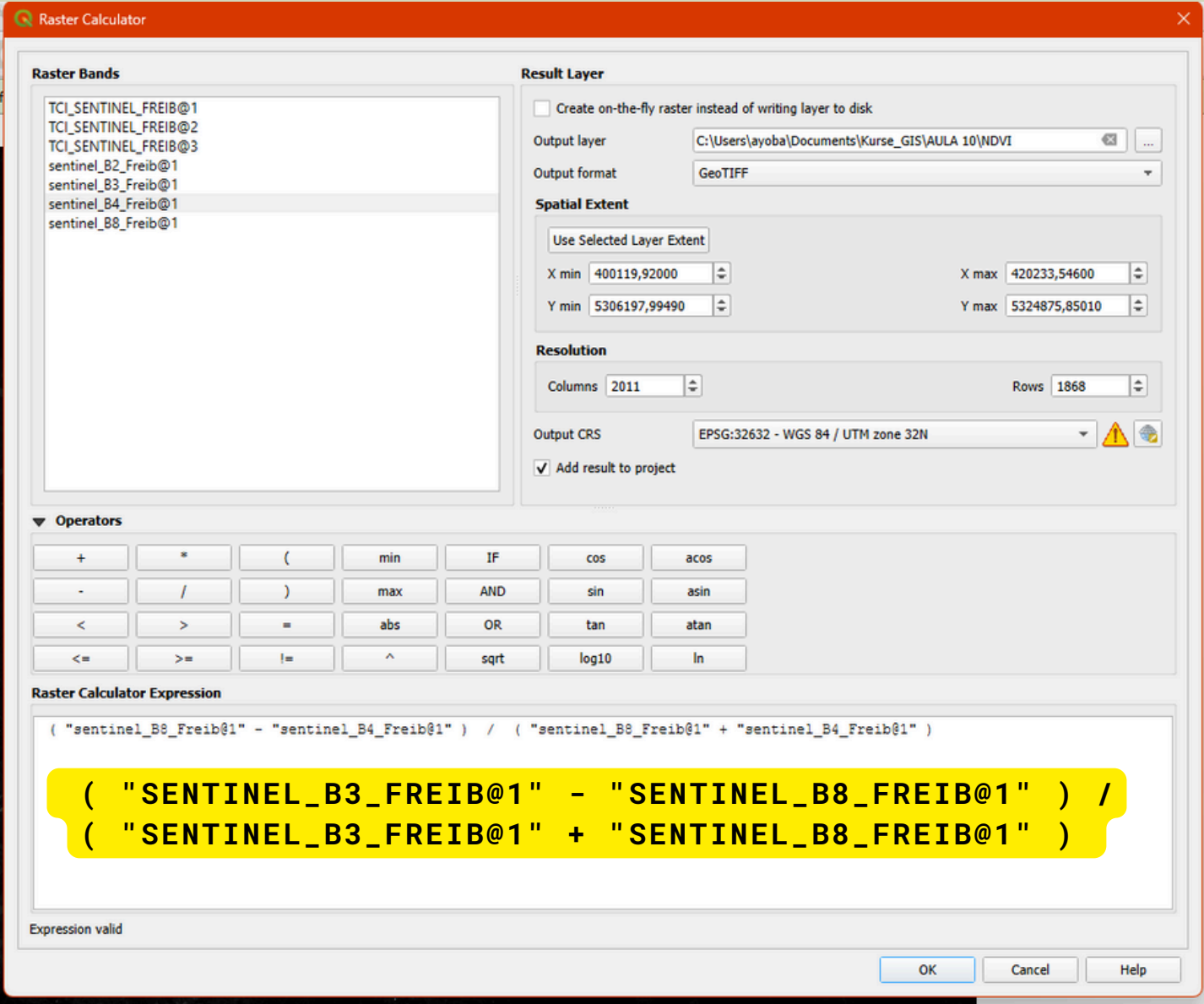
4 – PROCESS NDWI IN QGIS

$$NDWI = \frac{\text{Green} - \text{NIR}}{\text{Green} + \text{NIR}}$$

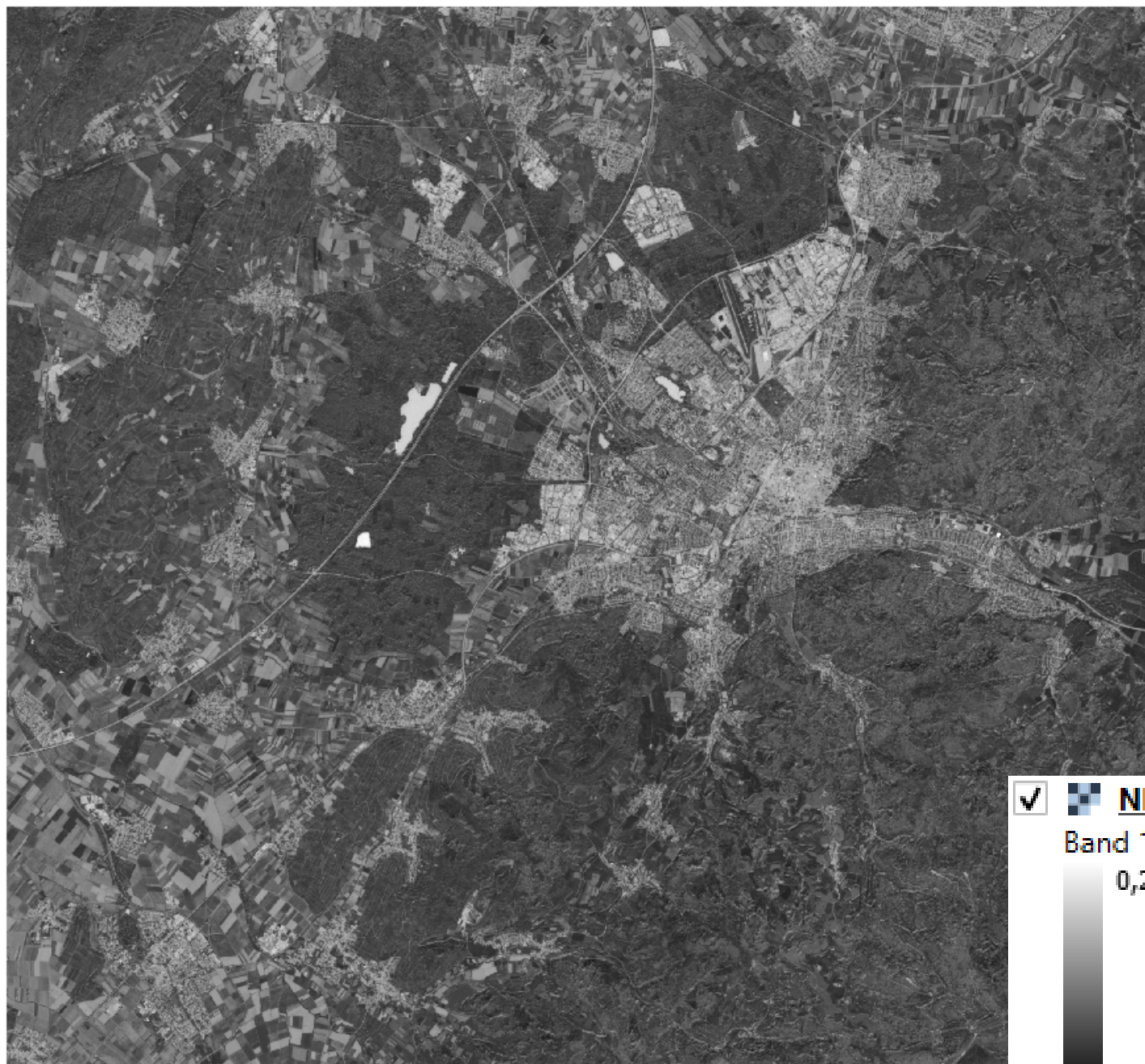
BAND	NAME
B02	Blue (Azul)
B03	Green (Verde)
B04	Red (Vermelho)
B08	NIR (Near Infrared)

$$NDVI = \frac{B3 - B8}{B3 + B8}$$

MENU RASTER → RASTER CALCULATOR



4 – PROCESS NDWI IN QGIS



NDWI value	Meaning
> 0.1	Water (strong indicator)
0 to 0.1	Wet soil / shallow water / shadows
< 0	Vegetation, soil, buildings

Water always appears with positive NDWI values.

LECTURE #11

Processing Raster Data



NOW THAT NDVI AND NDWI ARE EASY...

YOU CAN TRY TO STUDY AND COMPUTE BY YOURSELF THE 'NORMALIZED DIFFERENCE BUILT-UP INDEX' (NDBI).

NOTE THAT THE NDBI USES THE SWIR BAND, WHICH HAS A 20 M SPATIAL RESOLUTION, MEANING YOUR FINAL INDEX WILL ALSO BE AT 20 M.

YOUR TASKS:

1. LEARN THE PURPOSE OF NDBI AND UNDERSTAND WHY IT IS USED TO IDENTIFY URBAN AREAS.
2. IN QGIS, LOCATE AND LOAD THE SENTINEL-2 BANDS NEEDED FOR THE INDEX:
 - NIR = BAND 08 (10 M)
 - SWIR = BAND 11 (20 M)

LECTURE #11

Processing Raster Data



5. COMPUTE THE NDBI USING THE RASTER CALCULATOR:

6. $NDBI = \frac{SWIR - NIR}{SWIR + NIR}$
 $NDBI = \frac{SWIR - NIR}{SWIR + NIR}$

7. VISUALIZE YOUR RESULT USING AN APPROPRIATE COLOR RAMP AND INTERPRET THE BUILT-UP AREAS IN YOUR MAP.

-